Early Prediction of Peaks in Time Series: Quantile Regression with Asymmetric Laplace Noise and a Data-Driven Peak-Region Construction

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Abstract

Anticipating the location and amplitude of a peak before it is fully observed is a recurring problem in environmental and public-policy time series: peaks of air pollutant concentrations, temperature extremes during heat waves, peak electricity demand, flood levels, and epidemic incidence peaks all require decisions to be taken *before* the turning point is observable. We model the observations as a Gaussian-shaped unimodal signal corrupted by additive noise and develop a quantile-based procedure for estimating its amplitude, location, and spread from partial observations. For fixed location and spread, the amplitude estimator is a weighted τ-quantile of a transformed version of the data, generalizing the least-absolute-deviation (LAD) estimator of Dermoune et al. (2026). We introduce a τ-aligned asymmetric Laplace noise model whose negative log-likelihood is proportional to the quantile check loss, providing a unified probabilistic framework in which the quantile level and the signal-to-noise ratio can be studied jointly. Because the sequential peak-location estimates need not stabilize early, we propose a two-stage construction based on the oscillations of the estimation trajectory that produces candidate entry and exit times, hence candidate peak regions, using only information available at the current observation horizon, without knowledge of the true peak. A Monte Carlo study assesses the empirical coverage of the resulting intervals, and the methodology is illustrated on daily COVID-19 case counts in China, France, and Germany during the first pandemic wave. The framework is directly transferable to environmental peak events such as pollution episodes and climate extremes.

Keywords: peak prediction; quantile regression; asymmetric Laplace noise; entry and exit times; extreme events; early warning; time series.

JEL classification: C22, C53, Q54, I18.

1 Motivation and research question

Many decisions in environmental management and public policy must be taken *before* the turning point of a crisis is observable: declaring pollution alerts while concentrations are still rising, activating heat-wave plans before the temperature maximum, dispatching spare generation before the electricity demand peak, releasing reservoirs before the flood crest, or scaling hospital capacity before the incidence peak of an outbreak. All share the same statistical structure: a noisy, unimodal time series whose peak location L and amplitude A must be inferred from observations $I(1), \ldots, I(T)$ taken before the peak, with possibly $T < L$.

Standard forecasting tools (ARIMA models, recurrent neural networks) describe temporal dependence but are not designed to estimate the timing and magnitude of a *future peak*; a model can forecast well on average yet locate the turning point poorly. We build on the peak-estimation framework of Dermoune et al. (2026), who model the signal as a Gaussian-shaped peak corrupted by Laplace noise and estimate its parameters by least absolute deviations (LAD, i.e., median regression), and extend it in three directions:

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1. **Quantile generalization.** The LAD criterion is replaced by a general quantile regression criterion indexed by $\tau \in (0,1)$. This is natural for peak events: overestimating and underestimating the peak time have asymmetric costs (acting too late is typically much costlier than acting too early), and environmental and incidence series often exhibit skewed noise.

2. **Probabilistic foundation.** We introduce a τ-aligned asymmetric Laplace (AL) noise model whose negative log-likelihood coincides with the quantile check loss. The criterion thereby acquires a likelihood interpretation, and the effects of τ and of the signal-to-noise ratio (SNR) can be studied within one framework.

3. **Operational peak-region detection.** When the sequential peak-location estimate does not stabilize early enough for a reliable point prediction, we construct candidate *entry and exit times* of the peak region from the oscillations of the estimation trajectory alone. The construction is fully operational, it never uses the true (future) peak location, and is validated by Monte Carlo simulation.

The methodology is illustrated on daily COVID-19 case counts in China, France, and Germany during the first wave of 2020, a canonical noisy unimodal series whose peak had to be anticipated for planning. Section 7 discusses the direct transferability of the framework to environmental peak series.

2 Model and quantile-based estimation

We observe a unimodal signal corrupted by additive noise,

$$I(t) = A \exp\left(-\frac{(t-L)^2}{S^2}\right) + e(t), \qquad t = 1, \dots, T, \tag{1}$$

with amplitude $A > 0$, peak location $L > 0$, spread $S > 0$, and noise $e(t)$. The decision problem is to estimate (A, L, S), and in particular to predict L, using only observations up to a horizon T that may precede the peak. Given a quantile level $\tau \in (0,1)$ and the check loss

$$\rho_\tau(u) = u\big(\tau - \mathbf{1}_{\{u<0\}}\big), \tag{2}$$

the parameters are estimated by minimizing the empirical quantile loss

$$f_\tau(T, a, l, s) = \frac{1}{T} \sum_{t=1}^{T} \rho_\tau\left(I(t) - a \exp\left(-\frac{(t-l)^2}{s^2}\right)\right) \tag{3}$$

over candidate locations l in a prescribed future horizon and scale values s, the amplitude being re-estimated for each pair (l, s).

2.1 Weighted quantile estimation of the amplitude

For fixed (l, s), define the weights $w(t) = \exp(-(t-l)^2/s^2)$ and the transformed data $x(t) = I(t)\exp((t-l)^2/s^2)$. Since $\rho_\tau(cu) = c\rho_\tau(u)$ for $c > 0$, minimizing (3) over a is equivalent to minimizing $\sum_t w(t)\rho_\tau(x(t)-a)$, whose solution is the weighted τ-quantile of the transformed observations:

Proposition 1. *Let $(x(t), w(t))_{t=1}^T$ be real values with $w(t) > 0$. For any $\tau \in (0,1)$, a minimizer of $a \mapsto \sum_{t=1}^T w(t)\rho_\tau(x(t)-a)$ is obtained as follows. Sort the $x(t)$ increasingly, $x(\pi(1)) \le \cdots \le x(\pi(T))$, and let $k = \min\{j : \sum_{i=1}^j w(\pi(i)) \ge \tau \sum_{i=1}^T w(i)\}$. Then $q_\tau = x(\pi(k))$.*

The estimator thus reduces to a one-dimensional profile: for each candidate location l, optimize s, compute the weighted quantile amplitude in closed form, and evaluate (3). At $\tau = 1/2$ the LAD estimator of Dermoune et al. (2026) is recovered, so the method is a strict generalization. The quantile formulation brings two practical advantages for peak monitoring: robustness to skewed noise, and the possibility of targeting conservative (upper-quantile) or aggressive (lower-quantile) peak estimates according to the relative costs of late versus early action.

3 Likelihood interpretation: τ-aligned asymmetric Laplace noise

To study the estimator probabilistically, we use the asymmetric Laplace density with mode m, scale $\lambda > 0$ and asymmetry $k > 0$,

$$f_{m,\lambda,k}(x) = \frac{\lambda}{k + k^{-1}} \begin{cases} \exp\Big(\frac{\lambda(x-m)}{k}\Big), & x < m, \\ \exp(-\lambda k(x-m)), & x \geq m, \end{cases}$$

for which $\mathbb{P}(X < m) = k^2/(1+k^2)$. Setting $m = 0$ and

$$k = \sqrt{\frac{\tau}{1-\tau}} \tag{4}$$

gives $\mathbb{P}(e(t) < 0) = \tau$: the noise is *aligned* with the quantile criterion, its τ-quantile being exactly zero. Equivalently, the aligned density is

$$f_{\tau,b}(x) = \frac{\tau(1-\tau)}{b} \exp\Big(-\frac{1}{b}\rho_\tau(x)\Big), \qquad b > 0, \tag{5}$$

since $\int e^{-\rho_\tau(x)/b} dx = b/[\tau(1-\tau)]$. Hence $-\log f_{\tau,b}(e) = \text{const} + \rho_\tau(e)/b$, so maximum likelihood estimation coincides exactly with minimization of the empirical quantile loss (3). The symmetric Laplace model of Dermoune et al. (2026) corresponds to $\tau = 1/2$. The variance of (5) is

$$\mathrm{Var}(e) = b^2 \, \frac{\tau^2 + (1-\tau)^2}{\tau^2(1-\tau)^2}, \tag{6}$$

which, with the convention $\mathrm{SNR} = A/\sqrt{\mathrm{Var}(e)}$, yields the scale calibration

$$b = \frac{A}{\mathrm{SNR}} \, \frac{\tau(1-\tau)}{\sqrt{\tau^2 + (1-\tau)^2}}. \tag{7}$$

Equations (4)–(7) define a coherent simulation design in which each quantile level is paired with noise having its τ-quantile at zero and a common SNR, so that the effects of τ and of the noise level can be disentangled.

4 Sequential estimation, stability, and the peak region

Because decisions are taken online, we study the whole estimation trajectory $T \mapsto \hat{l}(T)$ as data accrue, using two complementary devices.

4.1 Retrospective stabilization time

For a tolerance $\varepsilon > 0$, define

$$T_{\mathrm{stab}} = \min\Big\{T : \; \max_{u \geq T} \hat{l}(u) - \min_{u \geq T} \hat{l}(u) \leq \varepsilon \Big\}. \tag{8}$$

Under τ-aligned AL noise with $A = 1$, $L = 22$, $S = 10$ and $\mathrm{SNR} = 40$, the stabilization times over $\tau \in \{0.25, 0.5, 0.75, 0.95\}$ are $17, 17, 11, 10$ respectively (with $\varepsilon = 4$): upper quantiles stabilize earlier in this design. We stress that (8) is a retrospective diagnostic; the construction below is its operational counterpart.

4.2 Oscillations and candidate entry–exit times

Fix a horizon h with $L < h$, and let $T < h$ be the current observation time. For $t \leq T$, define the retrospective oscillation of the estimation path,

$$\omega_l(t, T) = \max_{t \leq t_1, t_2 \leq T} \big|\hat{l}(t_1) - \hat{l}(t_2)\big|. \tag{9}$$

For fixed T, $t \mapsto \omega_l(t,T)$ is nonincreasing; its strict decreases define candidate exit times $T_{\text{exit}}^{(1)}(T) > T_{\text{exit}}^{(2)}(T) > \cdots$ with levels $\epsilon_k'(T) = \omega_l(T_{\text{exit}}^{(k)}(T), T)$. For each candidate exit time, the Stage-2 set

$$E_k(T) = \Big\{ t < T_{\text{exit}}^{(k)}(T) : \omega_l\big(t, T_{\text{exit}}^{(k)}(T)\big) \geq \epsilon_k'(T) \Big\} \tag{10}$$

yields $\epsilon_k(T) = \omega_l(T_k^*(T), T_{\text{exit}}^{(k)}(T))$ with $T_k^*(T) = \max E_k(T)$, and the entry time

$$T_{\text{entry}}^{(k)}(T) = \min\Big\{ t < T_{\text{exit}}^{(k)}(T) : \omega_l\big(t, T_{\text{exit}}^{(k)}(T)\big) = \epsilon_k(T) \Big\}. \tag{11}$$

Each candidate k defines a peak region $\mathcal{R}_k(T) = [T_{\text{entry}}^{(k)}(T), T_{\text{exit}}^{(k)}(T)]$ of width $w_k(T) = T_{\text{exit}}^{(k)}(T) - T_{\text{entry}}^{(k)}(T)$. All endpoints in (9)–(11) depend only on the estimation trajectory available at T; the true peak location L is never used.

4.3 Operational selection of an interval

Since $\hat{l}(T)$ is available at horizon T while L is not, we select the narrowest candidate containing the current estimate:

$$\widehat{k}(T) = \operatorname*{arg\,min}_{k \in \mathcal{K}(T)} w_k(T), \qquad \mathcal{K}(T) = \Big\{ k : T_{\text{entry}}^{(k)}(T) \leq \hat{l}(T) \leq T_{\text{exit}}^{(k)}(T) \Big\}, \tag{12}$$

yielding the data-driven interval $\widehat{\mathcal{I}}(T)$. The first selection horizon $T_{\text{est}} = \min\{T : \mathcal{K}(T) \neq \varnothing\}$ is an operational stopping time. We emphasize that $\widehat{\mathcal{I}}(T)$ is an interval estimate whose construction alone carries no coverage guarantee; coverage must be assessed empirically, as we do next.

5 Monte Carlo evidence

We generate $M = 1000$ independent replicates of (1) with $A = 1$, $L = 22$, $S = 10$, $h = 30$, τ-aligned AL noise calibrated by (7), and SNR $\in \{1000, 100, 1\}$. For each replicate we apply the selection rule (12) at every horizon and record the interval-selection rate and the empirical coverage of L among selected intervals (Table 1).

At SNR = 1000 and 100, no interval is selected before $T = 22$; thereafter the selection rate and empirical coverage are both equal to one for all quantile levels. At SNR = 1, intervals may be selected early but none contains $L = 22$ before the peak; at $T = 30$ the empirical coverage among selected intervals is 0.613, 0.565, 0.637 and 0.844 for $\tau = 0.25, 0.5, 0.75, 0.95$, with selection rates above 0.996. Thus, in medium-to-high SNR regimes the procedure is conservative: it remains silent until the signal is informative and then covers the peak with probability one. Under very low SNR it still selects intervals frequently, but selection no longer implies coverage, an honest limitation, arguing for pairing the interval with an SNR diagnostic rather than treating it as a confidence interval.

6 Empirical illustration: COVID-19 incidence peaks

We apply the methodology to daily confirmed COVID-19 cases during the first wave of 2020 in China (Jan 22–Mar 22; JHU CSSE via the `COVID19` R package), France (Jan 15–May 31; ECDC), and Germany (Jan 28–Jul 5; ECDC), for $\tau \in \{0.25, 0.5, 0.75, 0.95\}$. These are canonical noisy unimodal trajectories whose peaks had to be anticipated for health-system planning, and they play the same methodological role as an environmental episode series.

For each country and quantile level we compute the estimation trajectory $T \mapsto \hat{l}(T)$ and evaluate the entry–exit construction of Section 4. Since the realized peaks are known retrospectively ($L = 22$ for China, $L = 78$ for France, $L = 61$ for Germany on the chosen time scales), candidates are assessed ex post by selecting, among peak regions strictly containing L, the one of minimal width. China yields narrow peak regions $(15, 26)$, $(15, 24)$ and $(13, 26)$ for $\tau = 0.25, 0.5, 0.95$ (no candidate contains L for $\tau = 0.75$); France yields $(72, 79)$, $(73, 82)$, $(70, 79)$ and $(70, 80)$; Germany yields the very sharp window $(60, 62)$ for $\tau = 0.25$ and $(56, 63)$ for $\tau = 0.95$. The results confirm that the estimated peak region depends on the quantile level, and that the construction behaves conservatively, remaining silent where

Table 1: Monte Carlo interval-selection rate, empirical coverage, and mean width at $T = 30$ ($M = 1000$ replicates).

τ	SNR	Selection rate	Coverage (selected)	Coverage (all)	Mean width
0.25	1000	1.000	1.000	1.000	18.70
0.25	100	1.000	1.000	1.000	12.66
0.25	1	0.998	0.613	0.612	5.62
0.50	1000	1.000	1.000	1.000	18.26
0.50	100	1.000	1.000	1.000	11.98
0.50	1	0.996	0.565	0.563	5.49
0.75	1000	1.000	1.000	1.000	18.79
0.75	100	1.000	1.000	1.000	12.72
0.75	1	0.996	0.637	0.634	5.30
0.95	1000	1.000	1.000	1.000	19.66
0.95	100	1.000	1.000	1.000	14.20
0.95	1	1.000	0.844	0.844	5.75

the trajectory is unstable, precisely the behavior desired in an early warning context.

7 Relevance to environmental econometrics

Although illustrated on epidemiological data, the framework is designed for the environmental applications emphasized by this conference. Peaks are the decisive feature of many environmental time series: daily maxima of $PM_{2.5}/NO_2$ during pollution episodes, extreme temperatures during heat waves, river discharge maxima, peak electricity demand during cold snaps, and price or load spikes in energy markets. In each case, (i) the series is unimodal around the episode; (ii) the noise is plausibly skewed (episodes build slowly and decay fast, or conversely), which the quantile criterion accommodates through τ; and (iii) actions (alerts, curfews, generation dispatch, reservoir releases) must be triggered before the peak, exactly the regime $T < L$ addressed here. The entry–exit construction provides an operational trigger (the first selection horizon T_{est}) together with an honest assessment of when the signal is too weak for a reliable peak region.

8 Conclusion

We have extended the LAD peak-estimation framework to general quantile levels, endowed it with a likelihood interpretation through τ-aligned asymmetric Laplace noise, and added an operational, fully data-driven construction of peak regions based on the oscillations of the sequential estimation trajectory. Monte Carlo evidence shows that the procedure is appropriately conservative when the signal is informative and honestly signals its own unreliability when it is not. The methodology is lightweight (profile optimization with a closed-form amplitude step), flexible in the noise distribution, and directly applicable to environmental peak events. Future work includes theoretical coverage guarantees for $\widehat{\mathcal{I}}(T)$, data-driven selection of τ via pinball-loss cross-validation, and multipeak and spatio-temporal extensions relevant for regional pollution and climate monitoring.

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